

ARPIT KUMAR

ML Engineer & AI Researcher | IIT Kharagpur | NLP · LLM/RAG · MLOps · Production ML

+91-9506035861 | arpitkumar@kgpian.iitkgp.ac.in | arpitkumar.dev | arpitkumar2004 | arpit-kumar-shivam | Lucknow, India

PROFESSIONAL SUMMARY

Final-year Dual Degree student at IIT Kharagpur (Chem Engg + AI and HPC Specialization, GPA 8.86), graduating April 2027 combining industrial engineering rigour with end-to-end ML expertise in predictive modelling, anomaly detection, data-driven analytics. Built and deployed production ML systems including an anomaly detection wellness engine (Isolation Forest, LOF) and multimodal prediction platform ranked Top 0.5% globally in Amazon ML Challenge 2025 (50K+ participants); engineered a Tri-Path RAG framework improving retrieval baselines by up to 19.8% across 10 benchmarks. Proficient in Python, Scikit-learn, NumPy, PyTorch, and MLflow across full data science lifecycle — from data cleaning and EDA to model deployment and monitoring. Seeking a Data Science graduate role at Schneider Electric to apply analytical and ML expertise to real-world energy management and operational efficiency challenges.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur | Kharagpur, India
Integrated B.Tech - M.Tech (Dual Degree) in Chemical Engineering
Specialization: Artificial Intelligence (AI) and High Performance Computing (HPC)

Jul 2022 - Apr 2027
GPA : 8.86/10

WORK EXPERIENCE

AI Researcher | Internship

May 2026 - Jul 2026

Sponsored Research & Industrial Consultancy (SRIC), IIT Kharagpur

On-site • Kharagpur, West Bengal, IN

- Built a **Tri-Path Multimodal RAG framework** for documents by combining dense BM25S retrieval, DuckDB Text-to-SQL, BLIP-2 vision retrieval, weighted RRF, cross-encoder reranking, and NLI attribution, by production-oriented Python architecture with scalable indexing and modular ingestion components.
- Improved retrieval and reasoning performance by integrating soft probabilistic routing with parallel text, table, and vision path, achieved **0.612 Recall@5**, **0.552 nDCG@5**, **0.578 SQL execution success**, and **0.657 attribution precision**, improved RAG baselines by **13.7–19.8% points** across benchmarks.
- Productionized the framework through PyPI packaging, GitHub Actions CI, automated vulnerability auditing, multi-version testing, and a **22-dimension evaluation harness**, validated the architecture across **10** benchmark datasets and mitigated **91%** of known RAG failure modes during robustness testing.

Technologies / Skills Used : PyTorch, Transformers, FastAPI, Qdrant, DuckDB, BM25s, Docling, PyPI

AI Researcher (UG) | Internship

May 2025 - Jul 2025

Advanced Technology Development Centre / Prof. Shyamal Kumar Das Mandal | IIT Kharagpur

On-site • Kharagpur, West Bengal, IN

- Developed two-phase **native language** identification framework using fixed Vision Transformer on L2 English speech, integrating Mel-spectrograms, F0 contours, and fluency descriptors to isolate acoustic and temporal cues across Hindi, Kannada, Malayalam, Tamil, and Telugu speakers in NISP datasets.
- Engineered **128-Mel spectrograms**, **log-F0**, and **14-feature** fluency representations from 44.1 kHz speech, trained ViT-small models with AdamW and cosine scheduling, implemented PyTorch feature caching to reduce epoch time from approximately **13 min. to 30 seconds** for efficient model training.
- Achieved **93.09% test accuracy** with Mel-spectrograms, outperforming **CNN** and **ResNet-18** baselines by **6.00%** and **4.80%**, stratified fluency analysis showed accuracy declining from **88.01%** at low fluency to **84.44%** at high fluency, revealing weaker accent cues in highly fluent speech overall.

Technologies / Skills Used : CNN (ResNet), BiLSTM, Mel-Spectrograms, MFCCs, t-SNE, Wav2Vec2, WavLM

PROJECTS

PrismPrice — Multimodal Product Price Intelligence Platform | Github Live Demo Website

Sep 2025 - Dec 2025

AI Engineer | Top 0.5%, Amazon ML Challenge'25 — national competition (sponsored by Amazon)

- Built a multimodal product-price prediction pipeline across **3 feature** modalities—**TF-IDF/SBERT text**, **CLIP/ResNet vision**, and unit signals, feeding **5 model** families with cross-validation and **OOF** stacking, plus persisted feature transformations, to create an end-to-end, production-ready pricing system.
- Ensured reproducible offline-to-online inference by packaging model weights, transformers, schema rules, and preprocessing into immutable run-scoped bundles, implemented versioned registry with staging/production promotion, deployment manifests, audit trails, and rollback support for governed model.
- Productionized real-time predictions through **FastAPI** endpoints, latency and error monitoring, and GitHub Actions automation, with integrated MLflow, DVC tracking, scheduled retraining, deployment validation, Hugging Face Spaces delivery with automated live health checks and model promotion gates.

Technologies / Tools Used : PyTorch, BERT, CLIP, Docker, Python, FastAPI, MLflow

LLM-Powered Employee Support Platform with RAG and Scalable AI Infrastructure | Github

Jan 2025 - Apr 2025

AI Engineer | 1st place, GC OpenSoft'25 — inter-hall software competition, IIT Kharagpur (sponsored by Deloitte)

- Reduced employee-support resolution time by **35%** by building an **LLM-powered** RAG platform with GPT-4o, LangChain, LangGraph, vector retrieval, and 5 specialized agents, enabling personalized employee analysis, context-aware recommendations, and faster assistance across diverse workplace support.
- Reduced hallucination rate by **40%** by designing a dual-knowledge RAG counseling pipeline combining employee records, psychological question banks, session-aware context, sentiment-driven branching, structured citation-based generation for reliable multi-turn conversations and automatic escalation.
- Boosted **API** throughput by **50%** by deploying **async FastAPI** services with concurrent agent execution, non-blocking **LLM communication**, optimized retrieval workflows, and production infrastructure using Docker, Kubernetes, and GCP, with autoscaling, load balancing, CI/CD, and real-time monitoring.

Technologies / Tools Used : FastAPI, Python, LangChain, OpenAI API, Groq, PyTorch, Docker, JWT

SKILLS

Programming Languages :	Python, C++, SQL, C, JavaScript
Frameworks & Libraries :	PyTorch, TensorFlow, Scikit-learn, FastAPI, Flask, JAX, Numpy, Pandas, Matplotlib, Seaborn, React
Data Science :	Machine Learning, Statistics, Probability, A/B Testing, Deep learning, Data Analysis, Data Visualization
Generative AI :	Retrieval-Augmented Generation, Model Interpretability, Distributed Inference, Quantization
Tools & Platforms :	MLflow, DVC, Git, GitHub, Docker, Dagshub, CI/CD, GitHub Actions
Databases :	MySQL, PostgreSQL

POSITION OF RESPONSIBILITY

Developers' Society, TSG, IIT Kharagpur

Advisor | Machine Learning & Software Architecture

Sep 2023 - Present
Kharagpur, West Bengal, IN

AWARDS & ACHIEVEMENTS

- Competitive Programming**: Achieved Expert rating on Codeforces (Peak: 1612), advance proficiency in data structures and algorithmic problem-solving
- Quant & ML Challenges**: Ranked Top 20 national in DTL Quant Challenge 2024 & Top 0.5% global Amazon ML Challenge 2025 out of 50K+ participant
- Mathematics**: Secured AIR 135 in the Integral Cup 2026 in 3 tracks Probability Theory & Statistics, Linear Algebra & Optimization and Integral Analysis
- Institute Championships**: Awarded 1 Gold Medal (General Championship Data Analytics 2025), 2 Silver Medals (Open IIT Data Analytics 2024 & Open IIT Case Study 2024), and 1 Bronze Medal (General Championship ChemQuest 2025) at IIT Kharagpur